

Peak Surfer



Capstone Project Guide

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ECE 3906 & ECE 4905
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2026-04-20

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1. Introduction

By their nature, wireless communications are vulnerable to jamming. One technique to mitigate the impact of a hostile jammer is “frequency hopping”, rapidly changing the frequency used for transmissions as a message is sent. Peak Surfer is a design for frequency hopping communications described by the NSA. Your job is to implement the Peak Surfer algorithm on a hardware platform. You will then create a formal report describing your implementation and providing recommendations for improvements or further experiments.

This paper is a project guide outlining the expectations of your capstone, along with providing a clear path to implementation. The Peak Surfer white paper provided by the NSA is a technical resource describing the algorithms, while this guide will describe a realistic system architecture that you will implement. This guide will also serve as a north star in offering guides, resources, and solutions to possible pitfalls you may encounter on your route.

The Peak Surfer project is an experiment. While you are expected to meet the outlined requirements, you may encounter fundamental design flaws or issues that prevent you from fulfilling said requirements. This is why the report component of the project is crucial; the NSA can make changes to the design and/or the capstone mentors can make changes to the architecture if necessary.

2. Scope & Goals

2.1. Scope of Work

The deliverables for this project are a working implementation of Peak Surfer on the provided radios and a report documenting the implementation. The radios will be able to receive and transmit data received from and provided by a host computer. The format and protection of this data is out of scope; you should only be concerned with communicating the given messages. Your report should contain a description of the implementation and its use, known issues or limitations, and suggestions for future work.

2.2. Requirements

The following are the minimum expectations of the completed implementation your capstone project produces. These requirements are supplemental to the ones outlined in the Peak Surfer white paper.

- The large block cipher should be Simon128/128.
- The radios should be able to operate without a host microcontroller (i.e. all Peak Surfer computations should occur on the radio's FPGA).
- The system should operate on the UHF band.
- The system should achieve a hop rate of at least 5k per second (one hop every 200 μ s) for 256 users (virtual users for which computations are performed as though they existed, not actual radios).

2.3. Stretch Goals

If your team finishes meeting the base requirements early, you can pursue any of the following project extensions. Failing to meet stretch goals will not negatively impact you in any way. You may design your own project extension that combines multiple goals or introduces new goals, however, your plan must be discussed with the capstone mentors and approved by the NSA.

2.3.1. Advanced Peak Surfer Implementation

1. The system should achieve a hop rate of at least 10k per second (one hop every 100 μ s) for 1024 users (virtual users for which computations are performed as though they existed, not actual radios).
2. Bring the power consumption of the Peak Surfer implementation (excluding the radio TX/RX power draw) below 2W

2.3.2. Advanced RF Implementation

Do any of the following:

- Implement a more sophisticated modulation scheme such as QAM-16 to increase throughput and/or reliability.
- Design and implement signal processing pipeline capable of removing basic environmental noise and interference.
- Demonstrate the system's effectiveness with more than two radios.

2.3.3. Advanced Networking

1. Demonstrate sending basic network packets over the system.
2. Design and implement a scheme that takes plaintext data, packages and encrypts it, before sending it over the Peak Surfer system. The receiver should handle the complementary pipeline for decryption. (This exceeds the scope of work, but will be a good learning experience.)
3. Add the ability to retransmit dropped packets and data error checking.
4. Make the data handling scheme as hardware based as possible so plain data can simply be passed in to the system and dispatched securely.

2.3.4. Red Teaming

1. Assess and research a realistic threat model for the Peak Surfer system, including both jamming and interception
2. Create or acquire an implementation of adversarial model(s) (one of which must include a point jammer)
3. Test your Peak Surfer implementation against the adversarial model(s).
4. If the your Peak Surfer implementation fails against any threat do any of the following: fix the implementation to handle such threats, provide a plan for implementation so it could be done in the future, or explain why it is not practical and/or desirable to fix.

2.3.5. Optimization

Do any of the following and compare it to your original implementation in terms of multiple metrics (e.g. Power consumption, logic elements, timing, etc.):

- Area optimize the algorithm implementations on the FPGA.
- Optimize all algorithms for latency and timing (including pipelining where applicable). Determine theoretical and realistic maximum of hops per second.
- Optimize the system for minimum power consumption (while still meeting the base requirements).

2.3.6. Hardware Design

1. Create an IC design that replaces the FPGA implementations. Account for the physical interfaces. Prioritize performance over area optimization.
2. Create a layout and perform some level of verification.
3. Tape out the design using a development kit of your choice.
4. Design a System-on-Module that incorporates your IC design and implements the Peak Surfer system.

3. Hardware

3.1. Description

3.2. Rationale

4. Roadmap

5. Appendix

5.1. Peak Surfer Project Description

5.2. HDL/Digital design Resources

5.3. Hardware Resources

5.4. Encryption References/resources

5.5. Additional Resources